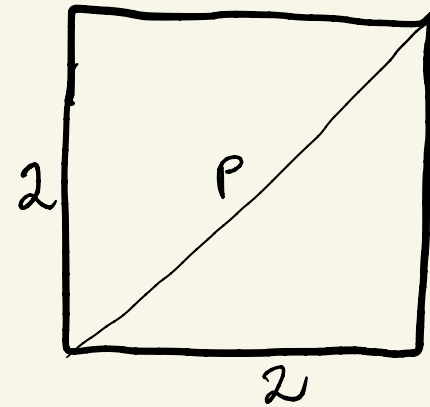


Example The equation $p^2 = 2$ is not satisfied by any $p \in \mathbb{Q}$.



Hippasus

Suppose we have a solution $p = \frac{m}{n}$, where m, n are coprime.

$$\left(\frac{m}{n}\right)^2 = 2 \iff m^2 = 2n^2$$

$\Rightarrow m^2$ is even.

$\Rightarrow m$ is even.

$\Rightarrow m^2$ is divisible by 4.

$\Rightarrow n^2$ is even.

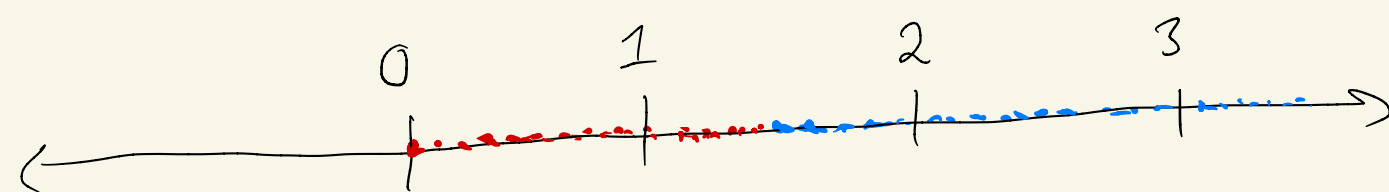
$\Rightarrow n$ is even.

However, m and n cannot be both even and coprime. Contradiction.

We must enlarge $\mathbb{Q}$ to have $\sqrt{2}$ and other things we want.

- Algebra solution: Algebraic extensions (see HW)
- Analysis solution: Completion

The rational numbers have "gaps"



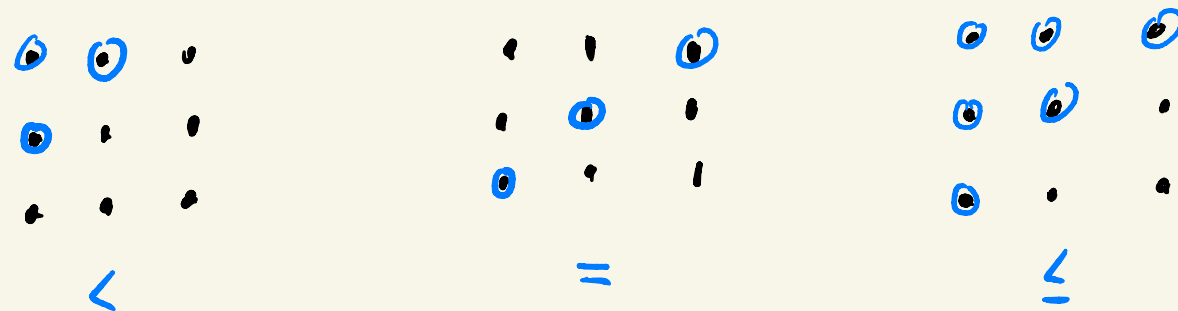
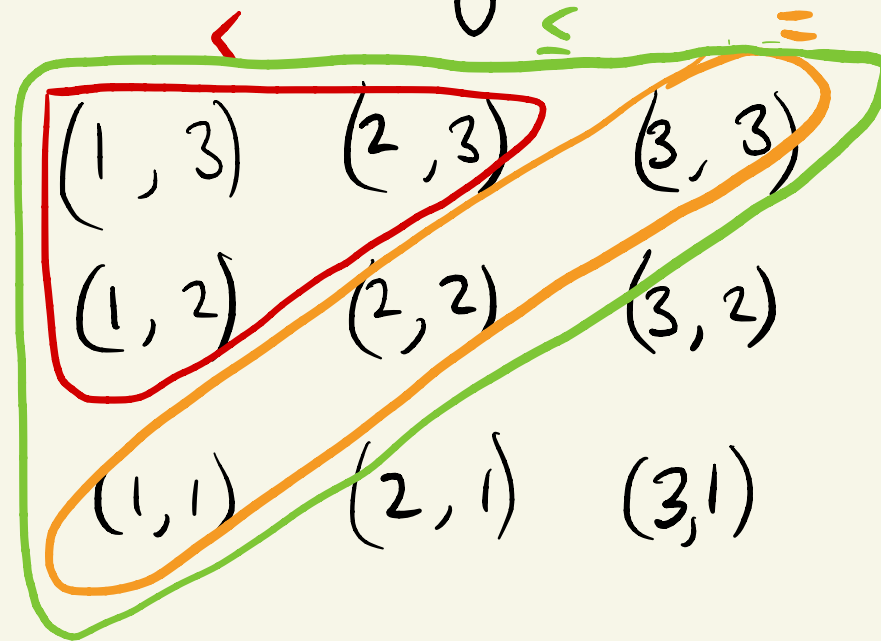
$$A = \{p \in \mathbb{Q}_{>0} \mid p^2 < 2\} \quad B = \{p \in \mathbb{Q}_{>0} \mid p^2 > 2\}$$

Even though between two rationals r, s there is always another rational, e.g. $\frac{r+s}{2}$.

Def A relation on a set S is a subset of $S \times S$.

Ex Some relations on the set $\underline{3} = \{1, 2, 3\}$.

Draw $\underline{3} \times \underline{3}$ as a grid:



Ordered Sets

Def Let S be a set. An order on S is a relation such that,

1) if $x \in S$ and $y \in S$ then one and only one of the following is true. $X < Y$, $X = Y$, $Y < X$,

2) If $x, y, z \in S$ if $x < y$ and $y < z$ then $x < z$ (transitivity).

Def The relations $\leq$, $>$, $\geq$ are defined from $<$.

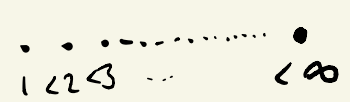
$$x \leq y \Leftrightarrow x < y \text{ or } x = y$$

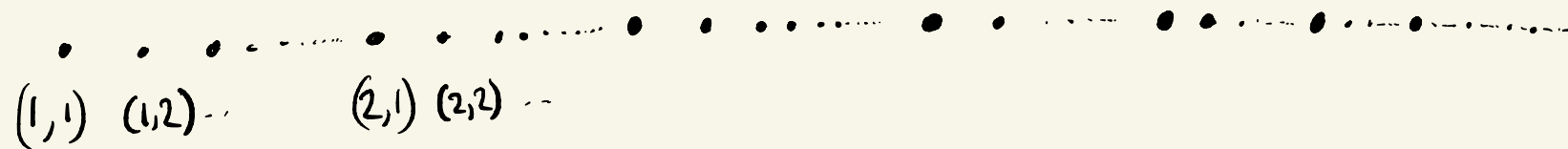
$$x > y \Leftrightarrow y < x$$

$$x \geq y \Leftrightarrow y \leq x$$

We call a set with a chosen order an "ordered set".

Examples

- $\mathbb{Q}$ with the standard order.
- $\mathbb{Z}, \mathbb{N}$, Any subset of an ordered set
- $\mathbb{N} \cup \{\infty\}$ 
- $\mathbb{N} \times \mathbb{N}$ with the lexicographic ordering:
 $(a, b) < (a', b')$ if $a < a'$ or $a = a'$ and $b < b'$



$(1,1) (1,2) \dots (2,1) (2,2) \dots$

Def An element x of an ordered set E is a maximum if $x \geq y$ for all $y \in E$.

Ex • $\max \{a \in \mathbb{Q} \mid a \leq 1\} = 1$

• $\max \{a \in \mathbb{Q} \mid a < 1\}$ does not exist

Def Let S be an ordered set. If there exists $\beta \in S$ such that $x \leq \beta$ for all $x \in E$ then E is bounded above, and β is an upper bound. Similarly, lower bound, bounded below.



If E is bounded above and below, it is bounded.

Ex • $\mathbb{N}$ is bounded below but not above in $\mathbb{Z}$.

• $\mathbb{N}$ is bounded in $\mathbb{Z} \cup \{\infty\}$.

• $\{p \in \mathbb{Q} \mid p < 1\}$ is bounded above.

Def Let $E \subset S$ be a subset of an ordered set.

β is the least upper bound, or supremum of E if

$\gamma < \beta$ implies γ is not an upper bound. We write $\beta = \sup E$.

Equiv: $\sup E = \min \{x \in S \mid x \text{ is an upper bound of } E\}$

Similarly, the infimum of E , or $\inf E$, is the greatest lower bound of E .

Example

1) If $A \subset S$ has a maximum $a \in A$, then $\sup A = a$.

2) $\sup \{r \in \mathbb{Q} \mid r < 1\} = 1$



$\sup$ is more robust version of $\max$.

Def An ordered set S has the least upper bound property if every subset which is bounded above has a supremum.

Spoilers: ① $\mathbb{Q}$ does NOT have the LUBP.

② $\{p \in \mathbb{Q}_{>0} \mid p^2 < 2\}$ has no supremum in $\mathbb{Q}$.

③ $\mathbb{R}$ will fix everything.

Fields

Def A field is a set F with the following structure:

- addition $+: F \times F \rightarrow F$
- multiplication $\cdot: F \times F \rightarrow F$
- additive inverse $F \rightarrow F$ denoted $x \mapsto -x$
- multiplicative inverse $F \rightarrow F$ denoted $x \mapsto \frac{1}{x}$
- $0 \in F$
- $1 \in F$

Satisfying the following axioms for all $x, y, z \in F$:

Addition

- commutativity $x + y = y + x$
- associativity $(x + y) + z = x + (y + z)$
- identity $x + 0 = x$
- inverse $x + (-x) = 0$

Multiplication

- commutativity $xy = yx$
- associativity $(xy)z = x(yz)$
- identity $1x = x$ if $x \neq 0$
- inverse $x \frac{1}{x} = 1$ if $x \neq 0$

Distributivity

- $x(y+z) = xy + xz$

Examples

- $\mathbb{Q}$ the rationals
- $\mathbb{Q}(\sqrt{3})$ see homework. "Algebraic extensions of $\mathbb{Q}$ ".
- $\mathbb{Z}/p$ where p is prime.
- $\mathbb{R}$ real numbers
- $\mathbb{C}$ complex numbers

Prop If x, y, z are elements of a field F , then

a) $x + y = x + z \Rightarrow y = z$

b) $x + y = x \Rightarrow y = 0$ (0 is unique)

c) $x + y = 0 \Rightarrow y = -x$ (negatives are unique)

d) $-(-x) = x$